FROM THE

EDITOR-IN-CHIEF

Seeing is believing. That old truism is no longer true. Just days ago, a video of Bollywood actress Kajol changing into an outfit did the usual heated rounds of social media before it was flagged as a 'deepfake'. Her face had been grafted onto the video of another person in so believable a fashion that most people would fall for it. Just before that came another deepfake which created a big brouhaha when one of the current divas of popular cinema, Rashmika Mandanna, went through the same ordeal. And the video went wildly viral before it was revealed that the body actually belonged to British-Indian social influencer Zara Patel. Such events threaten to become emblematic of our age. Luckily, they left no lasting scars, but the episodes did point to an area of darkness in our midst that's expanding at perilous speed. So much so that even Prime Minister Narendra Modi underlined the concern, saying he had seen a deepfake video of him doing a garba dance, something he hadn't done since his school days. That it turned out to be a 'normal fake' with a real-life lookalike does not ease the alarm.

The things you can do with advanced digital tools are mind-boggling and frightening at the same time. As the ruling deity of our times, the internet is omnipresent and intimate, but not necessarily a benevolent deity. We have seen previously unknown levels of nasty, brutish behaviour in public spaces, and fake news is already such a thriving industry that fact-checking has become an ancillary profession. Now, with the arrival of Artificial Intelligence (AI), a rash of new apps puts the means to do high-end digital sorcery at the disposal of ordinary people. 'Deepfake' is a live, ticking bomb in every hand that holds a smartphone. But what is it exactly? It's manipulating reality digitally from available data, both audio and visual, with such finesse that you can't tell the difference between real and artificial. The morphing of images has been around for some time, but deep learning-based AI mated with the urge for misinformation has taken things to an unimaginable level. AI can study real-world phenomena in such granular detail that it can then simulate them nearly exactly. The visual content you get from that is a weapon of mass deception.

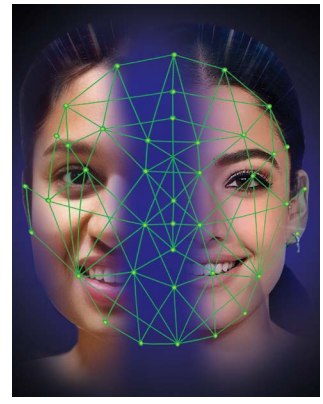


Illustration by NILANJAN DAS

The sinister uses it can be put to is devastating in its efficacy and scope. Deepfakes have the potential to not only damage individual reputations but disrupt the whole fabric of society. It is already being used for generating celebrity pornographic material, fake news, and for bullying and financial fraud. Public figures are the most vulnerable. One of the biggest dangers in a country like India, where literacy levels are challenging, is that they could determine the outcome of elections. It is so easy to create false narratives and change the public discourse because the electorate can be easily deceived. Recently, a morphed video of a *Kaun Banega Crorepati* clip went viral, with the intention of putting Madhya Pradesh chief minister Shivraj Singh Chouhan in a very unflattering light, just days before voting in the state. At an even less abstract level, as we have seen in Gaza, it can govern the conduct of wars. Both sides in the Israel-Hamas conflict have been guilty of weaponising deepfakes to sway world opinion. It's tragic proof that tens of thousands of lives can be at stake. In India, with all its mini-conflicts, imagine the amount of social anarchy that can be unleashed. If information is war, this is the smog of war.

There is an imperative need to regulate this phenomenon. The first recourse is technological. We need sophisticated content-verification tools to detect fakes. Fortunately, the technology that creates deepfakes can also, for the most part, recognise them. Before images are created by deep generative tools, real-world images are scrutinised to understand their formal 'architectural' detail. The trouble is perpetrators of deepfakes have access to the same tools, which makes detection far more challenging. And unfortunately, as technology has become so democratised, there are far more potential creators of deepfakes out there than there are detectors. One other possibility being contemplated is digital watermarking. Content lacking a watermark would be an alert for fake content.

The other critical intervention can come from the law. Only the fear of punishment can create a sufficient deterrent force. The government must provide a regulatory framework that lays down rules like watermarking and severe penalties for violation and misuse. In an interview with Group Editorial Director Raj Chengappa, the Union Minister of State for Electronics, Information Technology Rajeev Chandrasekhar spoke of how thinking is evolving around this issue in the Modi government. Already, the IT Act of 2000 has gone through three key revisions in the past three years that make the 'Safe Harbour' clause conditional for platforms like Facebook, YouTube et al. Once objectionable content is flagged, they have to take it down within a specified time-frame or be liable for prosecution. But the minister says it's time for a whole new law. The Digital India Act, going through multiple levels of review, will replace the 2000-vintage law. "When the IT Act came into being, the internet was a very innocent place. Techno-optimism was the norm," says Chandrasekhar. The new law will be "much more contemporary, much more aligned to deal with these modern challenges...the diversity and the complexity". It's also a calibrated approach that does not wish to disable a whole sunrise sector just for the perils that lurk in its darker corners. He foresees a "nimble, fleet-footed" law that is "pro-innovation, pro-user and pro-citizen safety". This can't happen soon enough.

In this week's cover story, Associate Editor Ajay Sukumaran maps the fast-developing technology that makes fakes more authentic day by day and the spread of its use. Faking itself is not new as we had Photoshop in the '90s where things like face swaps could easily be done. Deep Learning techniques have changed the game dramatically. We are likely to face a tsunami of deepfakes very soon. This is the flip side of digital India. Since mobile phones have become so universal, fakes go viral like a pandemic and cause far-reaching damage. That is why we are calling it 'Clear and Present Danger'. We live today in a post-truth world. In fact, truth is fast becoming an endangered species with all kinds of false narratives being promoted on social media. The proliferation of deepfakes makes getting to the truth very onerous. If you don't know it's fake, will you ever know the Truth?

(Aroon Purie)